

AC-Bench: Diagnosing Text-Induced Shifts in Visual Emotion Judgments

Anonymous submission

Abstract

Visual emotion analysis (VEA) aims to infer the affect elicited by visual content. However, in real-world image-text inputs, the two modalities do not always convey compatible affective information and may even provide competing affective cues. Existing multimodal affect evaluations typically assume emotional congruence between images and text, making it difficult to characterize how competing text alters models' judgments of visual affect. To address this gap, we introduce AC-Bench, a benchmark comprising 13,366 instances across nine controlled settings. AC-Bench uses controlled textual frames and provides human reference judgments for every instance under image-only, text-only, and joint image-text conditions. Evaluations across 12 vision-language models (VLMs) show that strong performance under image-only and modality-aligned conditions does not imply human-like cross-modal integration. We find that decision shifts induced by competing text differ significantly from those observed in humans in both direction and magnitude. Further representation analyses show that Visual and Text decisions correspond to separable internal states in middle-to-late layers, while lexical and attention evidence accounts for only part of this separation. Finally, a unified evaluation of activation-based intervention methods shows that they can reverse text-induced decision shifts only for specific models and under particular conditions, with effectiveness varying substantially across models.

Introduction

Visual Emotion Analysis (VEA) aims to infer the emotions evoked by visual content, supporting social-media understanding, affect-aware retrieval, and multimodal interaction (Borth et al. 2013; Zhao et al. 2016; Yang et al. 2023). In practice, images often appear with captions, titles, or comments that may provide useful context but may also impose interpretations unsupported by the image. Because visual affect is subjective and context-dependent (Achlioptas et al. 2023; Barrett, Mesquita, and Gendron 2011; Gendron et al. 2012; Wu et al. 2012), reliable systems should neither ignore language nor accept it uncritically. Recent VLMs show strong affect recognition and reasoning abilities (Bhattacharyya and Wang 2025; Lei et al. 2025; Cheng et al. 2024), yet may over-rely on text when modalities conflict (Deng et al. 2025; Zhang et al. 2025b; Nguyen et al. 2025). As illustrated in Fig. 1, a model may agree with humans on an image alone but follow a distracting textual frame in a case where the human judg-

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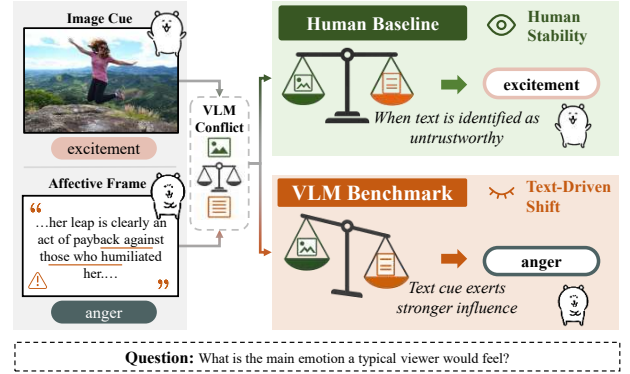


Figure 1: **AC-Bench isolates human-discordant affective shifts induced by Subjective Framing (SF).** The image supports *excitement*, whereas the accompanying text preserves the observable scene but adds a visually unsupported revenge interpretation targeting *anger*. The human panel discounts this framing and retains the image-side reference, while the VLM follows the textual target.

ments remain stable. Image-only accuracy therefore does not establish robustness to textual interference.

Existing benchmarks do not directly measure this failure mode. Image-centric VEA evaluates predictions against image-only labels (Yang et al. 2023; Bhattacharyya and Wang 2025), whereas naturally occurring image-text datasets entangle image content, textual function, emotion, and cue strength (Mohammad and Kiritchenko 2018; You, Jin, and Luo 2016). Psychological evidence motivates controlled, within-image manipulations (Gendron et al. 2012; Wu et al. 2012), while general cross-modal conflict benchmarks mainly study visually verifiable contradictions in objects, attributes, or events (Deng et al. 2025; Popordanoska, Li, and Blaschko 2026). Moreover, these evaluations rarely include human judgments matched to model conditions, making it difficult to characterize reasonable contextual integration and model-specific text following under the same manipulations. We therefore ask: *On a complete outcome-unfiltered cohort, how do VLM and human response destinations vary across controlled text conditions; among oracle-identified human-*

discordant Text decisions, how reversible are the model outputs?

To answer these questions, we introduce **AC-Bench**, a controlled, human-validated image–text benchmark built from randomly sampled EmoSet, social-media, and photographic images (Yang et al. 2023). It instantiates three routes of linguistic influence: **Affective Assertion (AA)** explicitly states an emotion (Gendron et al. 2012); **Perceptual Evidence (PE)** alters visually verifiable affective evidence (Machajdik and Hanbury 2010; Borth et al. 2013; Yang et al. 2023); and **Subjective Framing (SF)** preserves observable facts while adding an insufficiently supported appraisal (Scherer 2019; Wu et al. 2012). All instances undergo automated and expert checks. Human grounding combines an outcome-unfiltered, disjoint 5 + 5 split-panel study on 600 images with full-coverage longitudinal annotations reserved for construction and oracle diagnostics.

Our main contributions are as follows:

- We formulate human-referenced affective redirection and distinguish image, text, and other response destinations.
- We introduce AC-Bench: 13,366 outcome-unfiltered instances spanning three linguistic functions, nine settings, and complementary human protocols.
- Evaluating 12 VLMs and four interventions reveals function-specific redirection, separable Visual–Text states, and a model-dependent recovery–preservation trade-off.

Related Work

Visual Affective Understanding. Visual affective understanding has progressed from low-level affective features and Adjective–Noun Pairs to deep visual representations and large-scale datasets (Machajdik and Hanbury 2010; Borth et al. 2013; Rao, Li, and Xu 2020; Zhang, He, and Lu 2020). EmoSet further supports categorical emotion recognition with rich emotion-related attributes (Yang et al. 2023), while recent VLM-based systems extend the task toward instruction following, affective explanation, and multimodal emotion reasoning (Xie et al. 2024; Cheng et al. 2024; Luo et al. 2024). These studies substantially advance affect recognition and interpretation, but typically evaluate isolated images or naturally compatible multimodal inputs rather than controlling how competing text redirects an otherwise recognizable visual affect.

VLM Modality Conflict. A parallel literature examines VLMs under modality conflict, including text over-reliance, uncertainty-dependent modality preference, visual counter-facts against memorized priors, and conflicts between visual input and internal knowledge (Deng et al. 2025; Hua, Yun, and Pavlick 2025; Zhang et al. 2025a; Golovanevsky et al. 2025; Ortu et al. 2025). Most such settings concern factually adjudicable discrepancies over objects, attributes, or world knowledge. AC-Bench instead studies text-induced affective redirection through Affective Assertion (AA), Perceptual Evidence conflict (PE), and Subjective Framing (SF). Its complete evaluation cohort retains both stable and revised human responses; a separately labeled post-hoc subset identifies items on which the human panel remains anchored to

the image-side reference.

Activation Intervention and Reversibility. Activation-level intervention offers a training-free means of testing whether conflict-induced decisions are reversible. Contrastive Activation Addition (CAA) computes a steering vector from the mean paired residual-stream activation difference (Rimsky et al. 2024); Pixels Versus Priors (PvP) contrasts last-token representations elicited by pixel- and prior-focused prompts over the same image (Golovanevsky et al. 2025); Activation Steering Decoding (ASD) contrastively combines logits obtained under opposite-signed hidden-state interventions (Su et al. 2025); and Inference-Time Intervention (ITI) shifts selected attention-head outputs along learned directions (Li et al. 2023). We do not propose another steering algorithm. Instead, AC-Bench provides a unified affective-conflict protocol for comparing whether these four interventions can return human-discordant text-following predictions to the visual reference without redirecting them to a third emotion.

AC-Bench

Figure 2 provides a standalone overview; below we formalize the quality gates and evaluation cohorts. The independent split-panel study is a separate inference branch and does not affect benchmark membership. Table 2 defines the cohorts; prompts, annotation interfaces, condition-allocation details, and complete accounting are in the supplementary material.

Benchmark Design

Target and metrics. Panel 2(a) operationalizes the benchmark target: controlled text competes with a pre-text visual reference through three distinct linguistic functions. During construction, five annotators first label each image alone; their modal label defines the benchmark reference y_i^v only for images with a majority emotion, while the controlled text has target y_i^t . This image-only gate restricts the benchmark to panel-consensual visual affect but does not use any later image–text response.

Let $\mathcal{D}_{\text{full}}$ contain every instance that passes the image and text construction gates and receives five joint judgments. No joint outcome is used to retain an item. Full-coverage model evaluation and descriptive panel analysis use the outcome-unfiltered conflict set

$$\mathcal{D}_{\text{full}}^- = \{i \in \mathcal{D}_{\text{full}} : y_i^t \neq y_i^v\}.$$

For destination $c \in \{v, t, o\}$, the five human indicators are first averaged within item and each item receives equal weight; each model contributes one indicator. We report Image Match Rate (IMR), Text Match Rate (TMR), and Other Emotion Rate (OER), which are mutually exclusive. The full-coverage joint annotations were supplied by the same longitudinal panel and are therefore used only for descriptive dataset analysis and explicitly oracle-defined intervention cohorts. Primary human updating estimates instead use $\mathcal{D}_{\text{ind}}$, on which a new image-only panel establishes $y_i^{v,\text{ind}}$ before a disjoint panel evaluates the assigned joint inputs. Any human-resistant cohort is an analysis-only branch and never a benchmark-retention rule or an independently estimated human baseline.

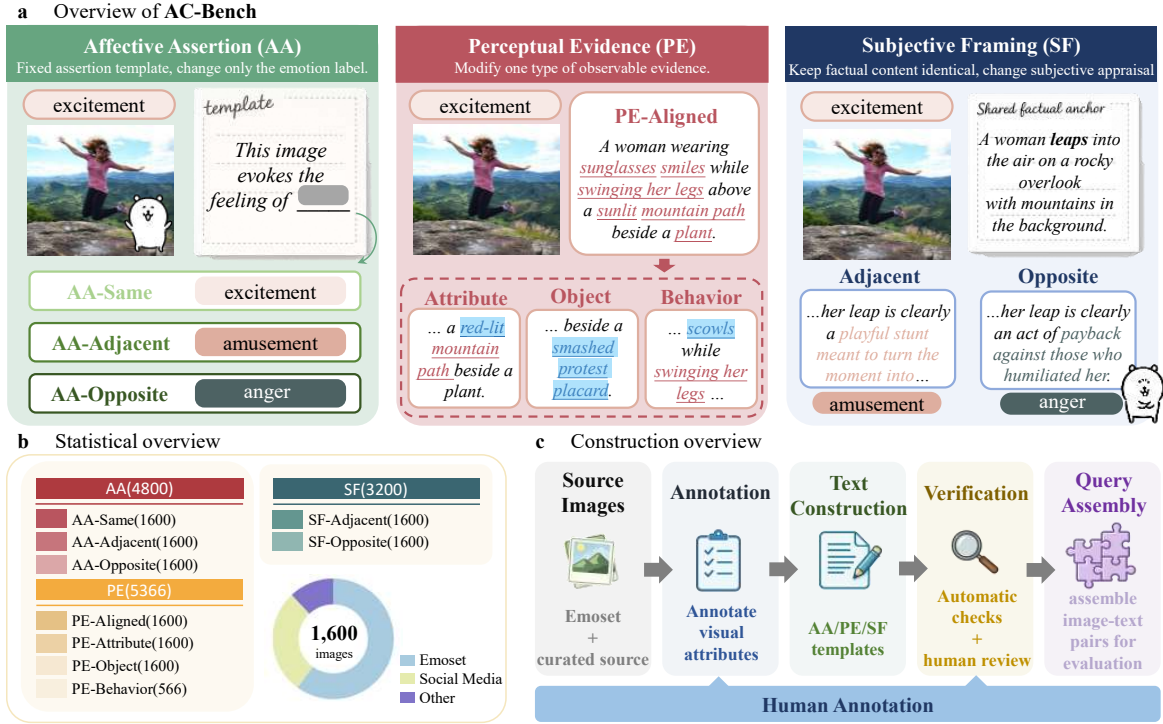


Figure 2: AC-Bench overview: controlled conflicts, composition, and construction. (a) Affective Assertion (AA) fixes the assertion template and varies only the emotion label across Aligned, Adjacent, and Opposite. Perceptual Evidence (PE) starts from a visually aligned factual description and changes one observable slot—Attribute, Object, or Behavior—while preserving the remainder. Subjective Framing (SF) preserves a shared factual anchor and varies only a visually unsupported appraisal to form matched Adjacent and Opposite frames. (b) The 1,600 images (60% EmoSet; 40% social-media and photographic sources) yield 13,366 instances across nine settings: 4,800 AA, 5,366 PE, and 3,200 SF. (c) The pipeline comprises source sampling, structured visual annotation, task-specific text construction, automatic checks and human review, and human-validated query assembly.

Affective Assertion (AA). AA fixes the template “This image evokes the feeling of ____.” and varies only a consistent, adjacent, or opposite emotion. Adjacent and opposite targets are the nearest non-identical and most distant categories under the polarity-aware wheel distance of Zhao et al. (2024), adapted from the eight-category scheme of Mikels et al. (2005). This isolates the influence of an explicit emotion concept.

Perceptual Evidence (PE). PE starts from an emotion-neutral factual description and changes exactly one visually checkable object, attribute, expression, or action while preserving the rest. It contains no explicit emotion label and tests whether the model resolves affective evidence from the image rather than a fluent counterfactual description (Machajdik and Hanbury 2010; Borth et al. 2013; Yang et al. 2023). Object and attribute conditions cover all 1,600 images; Behavior uses the 566 images with verified visible-person content.

Subjective Framing (SF). SF preserves observable facts and avoids emotion labels, but adds one visually unsupported appraisal of mental state, goal, agency, or consequence that humans recognize as an inappropriate basis for judging im-

age emotion (Scherer 2019; Wu et al. 2012). For example, it may retain a group’s smiles yet describe them as hollow or merely performative. This does not contradict the smiles, but adds an ungrounded interpretation. SF is retained when blind reviewers recognize the target and unsupported over-interpretation while confirming fact preservation; the later joint human emotion is measured as an outcome, not used as a construction gate.

Dataset Construction and Human Validation

Images and candidate validation. As summarized in Fig. 2(c), we draw candidates at random from EmoSet (60%) (Yang et al. 2023) and supplementary social-media and photographic pools (40%), without VLM pre-screening, then deduplicate and safety-check them. Before text, five annotators label each candidate; retaining cases whose modal emotion receives more than half the votes yields 1,600 images, with the mode used as y_i^v . AA is deterministic, whereas PE and SF are generated as matched single-slot edits. Automatic checks cover format, duplication, safety, lexical leakage, and pair structure. Five additional reviewers, blinded to target and condition relation, independently verify every PE

Model	AA			PE			SF		
	IMR↑	TMR↓	OER↓	IMR↑	TMR↓	OER↓	IMR↑	TMR↓	OER↓
<i>Open-source VLMs</i>									
Qwen2.5-VL-7B	25.31 (-48.0)	66.87	7.81	3.58 (-71.8)	61.95	34.46	15.97 (-46.6)	72.53	11.50
Qwen3-VL-8B	28.63 (-44.7)	62.09	9.28	12.03 (-63.3)	48.65	39.31	33.28 (-29.3)	48.75	17.97
LLaVA-NeXT-8B	9.37 (-63.9)	89.12	1.50	8.59 (-66.8)	46.35	45.06	23.10 (-39.5)	58.69	18.22
InternVL3.5-8B	16.18 (-57.1)	77.56	6.25	4.64 (-70.7)	53.43	41.93	18.69 (-43.9)	67.13	14.19
Phi-3.5	2.12 (-71.2)	97.81	0.06	3.83 (-71.5)	47.81	48.36	13.10 (-49.5)	72.69	14.22
Kimi-VL-A3B	18.94 (-54.4)	75.97	5.09	7.28 (-68.1)	55.67	37.05	21.22 (-41.4)	65.97	12.81
Step3-VL-10B	23.97 (-49.3)	67.65	8.38	8.75 (-66.6)	41.82	49.44	20.69 (-41.9)	67.62	11.69
Ovis2.5-9B	57.63 (-15.7)	34.03	8.34	10.40 (-65.0)	48.15	41.45	26.44 (-36.1)	61.53	12.03
MOSS-VL	3.16 (-70.1)	96.50	0.34	9.30 (-66.1)	46.66	44.05	19.72 (-42.9)	65.44	14.85
<i>Proprietary VLMs</i>									
GPT-4o	59.44 (-13.9)	28.82	11.76	50.13 (-25.2)	19.41	30.46	41.66 (-20.9)	45.59	12.75
GPT-5	<u>60.82</u> (-12.5)	29.00	10.19	<u>55.20</u> (-20.2)	16.50	28.31	47.47 (-15.1)	39.56	12.97
Gemini-3.0	63.05 (-10.3)	22.06	14.90	55.22 (-20.1)	13.25	31.52	<u>47.31</u> (-15.3)	40.63	12.07
Full-coverage panel (desc.)	73.30	19.10	7.60	75.37	7.92	16.71	62.58	27.33	10.08

Table 1: VLM and full-coverage longitudinal-panel response distributions (%) on the complete, outcome-unfiltered matched evaluation cohort; no item is selected by its joint human response. Human entries average the five response indicators per item, whereas each model contributes one decoded label per item. AA and SF are macro-averaged over Adjacent/Opposite; PE uses the 566-image subset shared by Object/Attribute/Behavior. Parentheses show the descriptive common-reference IMR difference $\text{IMR}_{\text{model}} - \text{IMR}_{\text{human}}$ on the same items; because y_v is panel-derived, this is not an independent estimate of differential updating. Bold/underline mark the top two VLM IMR values; humans are not ranked.

Cohort	Definition	Role
$\mathcal{D}_{\text{full}}$	Passes construction gates	Benchmark membership
$\mathcal{D}_{\text{full}}^-$	$y_i^t \neq y_i^v$; no response filter	Descriptive comparison
$\mathcal{D}_{\text{ind}}^{\text{pri},-}$	Disjoint panels; primary conflict cells	Confirmatory human-VLM
$\mathcal{T}_{H,m}$	$p_i^H \geq 4/5$ and model follows y_i^t	Post-hoc reversibility

Table 2: **Cohort definitions and inferential roles.** Only construction gates determine benchmark membership. The independent branch contains 480 primary images and 120 agreement-only anchors; only primary conflict cells enter the confirmatory comparison.

and SF candidate. Any hard constraint violation triggers regeneration; target recognition requires a majority, and mean naturalness must reach 3.5/5. SF additionally requires all five reviewers to confirm fact preservation, a single unverifiable appraisal, and no explicit emotion word. Full prompts and review criteria are in the supplementary material.

PE review checks that exactly the designated object, attribute, expression, or action conflicts with the image while all non-target facts remain fixed. SF review separately checks factual consistency, target recognizability, visual support, interference recognizability, and naturalness. This conjunctive gate prevents a merely ambiguous narrative from being labeled as subjective interference and keeps PE’s factual counterfactual distinct from SF’s fact-preserving interpretation.

Full-coverage annotation panel. The construction panel supplies image-only labels and five judgments for every retained joint instance in week-separated, randomized rounds. Annotators first choose an emotion and then its source; agreement is $\kappa = 0.599\text{--}0.651$ and $\alpha = 0.600\text{--}0.649$. No joint response removes an item. Because this panel supplies both y_v and later joint responses, its results are used only descriptively and for oracle diagnostics; human updating is estimated below.

Independent split-panel human study. Without consulting joint or model outcomes, five image-only annotators label 720 unseen candidates and freeze 600 by a prespecified majority gate. Five disjoint joint-only annotators then see each source image once: 480 images use balanced five-setting assignments and 120 serve as agreement anchors. VLMs are evaluated on the same primary cells. We denote the primary subset by $\mathcal{D}_{\text{ind}}^{\text{pri}}$; confirmatory destination rates use its conflict cells, $\mathcal{D}_{\text{ind}}^{\text{pri},-}$. The complete allocation and cross-fitted checks are in the supplement.

Dataset Characteristics

AC-Bench contains 13,366 instances across nine settings: 4,800 AA, 5,366 PE (three 1,600-image conditions plus 566 PE-Behavior items), and 3,200 SF. Every instance has five full-coverage longitudinal judgments, which remain secondary for human-updating inference. Image-only judgments recover the source emotion in 79.59% of images, text-only judgments recover the target in 92.68% of texts, and 91.43% of submitted generated candidates pass all condition-specific gates. Figure 2(b) summarizes dataset composition; Table 2

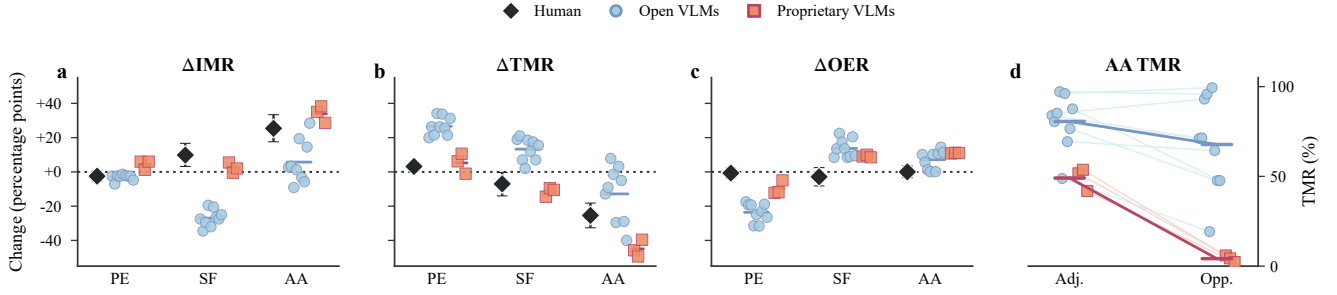


Figure 3: **Human and VLM condition contrasts differ in direction, destination, and compatibility sensitivity.** (a–c) Within-cohort changes in IMR, TMR, and OER. Black diamonds denote the longitudinal human panel with 95% bootstrap confidence intervals; blue circles and coral squares denote open and proprietary VLMs, and horizontal lines mark family means. PE is Behavior minus Object; SF and AA are Opposite minus Adjacent. (d) Full-cohort AA TMR for all 12 VLMs; thin lines pair Adjacent and Opposite within each model, and thick bars mark open/proprietary means. All estimates precede any human-resistance filter and are reported as secondary descriptive contrasts; the independent-panel estimand is defined separately.

separates benchmark membership from the independent and diagnostic branches.

Response Destinations of VLMs and Humans

Evaluation Setup

We use a fixed eight-way prompt and report Image Match Rate (IMR), Text Match Rate (TMR), and Other Emotion Rate (OER), which sum to 100%. We evaluate nine open VLMs—Qwen2.5-VL-7B (Bai et al. 2025), Qwen3-VL-8B (Bai, Cai et al. 2025), LLaVA-NeXT-8B (LMMS-Lab 2024), InternVL3.5-8B (Wang, Gao et al. 2025), Phi-3.5 (Microsoft 2024), Kimi-VL-A3B (Kimi Team 2025), Step3-VL-10B (Huang, Yao et al. 2026), Ovis2.5-9B (Lu et al. 2025), and MOSS-VL (OpenMOSS Team 2026)—plus GPT-4o (OpenAI, Hurst et al. 2024), GPT-5 (OpenAI 2025), and Gemini-3.0 (Google DeepMind 2025). We decode one emotion label per item; prompts and condition-level results are in the supplementary material. The proprietary models reach 95.74–98.13% on AA-Same, so their conflict-condition behavior is not attributable to failure on the aligned assertion template.

The 63.1/55.2/47.5% IMRs and 10.3/20.1/15.1-point deficits are descriptive full-cohort values, not confirmatory estimates. They correspond to Gemini-3.0 on AA and PE, and GPT-5 on SF, each compared with the full-coverage longitudinal panel in Table 1. Confirmatory comparisons instead use conflict cells in the 480-image primary subset $\mathcal{D}_{\text{ind}}^{\text{pri}}$, its independently established reference, and exactly the cells assigned to H_{joint} .

Confirmatory comparison. Table 3 reports the paper’s only confirmatory human–VLM estimand, comparing humans and macro-averaged VLMs on identical primary conflict cells. VLMs show reliably lower IMR and higher TMR across all functions; only PE also shows a reliable OER excess (+18.89 points). Thus, AA and SF chiefly redirect decisions from image to text, whereas PE also redirects them beyond both references. Table 1 and Fig. 3 are full-cohort descriptive, rather than paired, comparisons.

Task	Source	IMR	TMR	OER
AA	Human	73.12	17.95	8.93
	12-VLM mean	31.24	63.33	5.43
	Δ (V–H)	–41.88 [†]	+45.38 [†]	–3.50
PE	Human	71.90	9.21	18.89
	12-VLM mean	20.10	42.12	37.78
	Δ (V–H)	–51.80 [†]	+32.91 [†]	+18.89 [†]
SF	Human	60.01	27.14	12.85
	12-VLM mean	27.86	57.11	15.03
	Δ (V–H)	–32.15 [†]	+29.97 [†]	+2.18

Table 3: **Confirmatory split-panel human–VLM comparison on primary conflict cells $\mathcal{D}_{\text{ind}}^{\text{pri}}$.** Human denotes H_{joint} ; 12-VLM mean is the equal-weight macro-average over models evaluated on identical cells. Δ is VLM minus Human, in percentage points. [†] indicates that the source-image bootstrap 95% CI excludes zero. Complete intervals are reported in the supplementary material; inference is conditional on the fixed suite of 12 models.

Longitudinal-panel condition sensitivity. AA-Adjacent preserves image-emotion polarity in 98.4% of responses but has greater pairwise disagreement than AA-Opposite (0.411 vs. 0.222). PE-Behavior likewise raises disagreement for seven of eight emotions despite only a 3.2-point TMR increase. We therefore treat these matched longitudinal responses as descriptive rather than invariant ground truth.

Descriptive Condition Contrasts

We compute within-cohort ΔIMR , ΔTMR , and ΔOER between prespecified text conditions. PE and SF use the same 566 and 200 source images for humans and VLMs. For AA, the available human estimate uses the 100-image longitudinal panel, whereas all VLM estimates use the complete 1,600-image cohort; the AA comparison is therefore descriptive rather than paired.

Figure 3 shows that condition sensitivity depends on lin-

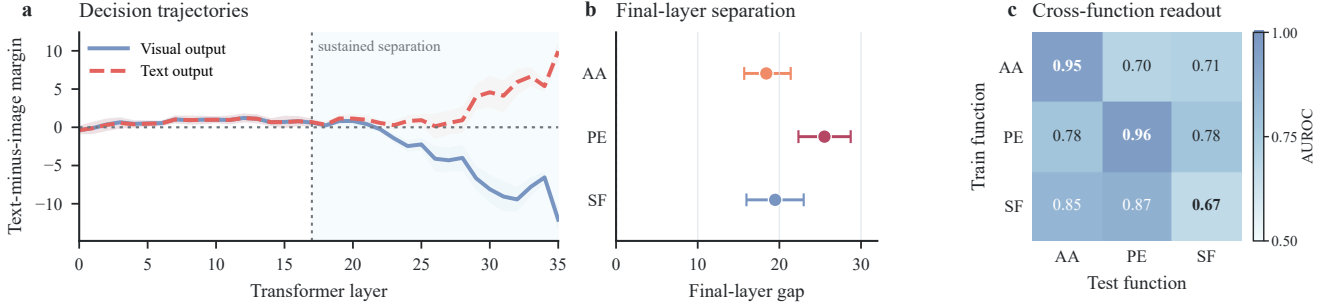


Figure 4: **Late-layer Visual-Text separation is shared across functions but does not reduce to a universal direction.** (a) Pooled trajectories for exactly matched Visual-Text pairs. (b) Function-specific final-layer gaps with 95% paired-bootstrap intervals. (c) Image-disjoint layer-23 train (row) to test (column) AUROC; bold diagonal cells are within-function, and label-only controls equal 0.50.

guistic function and response destination. From PE-Object to PE-Behavior, human TMR rises only 3.2 points (95% CI [1.4, 5.0]), whereas mean VLM TMR rises 21.3 and OER falls 20.1, with little IMR change (−1.2). From adjacent to opposite SF, humans move toward the image (+9.8 IMR, −7.0 TMR), while open VLMs move oppositely (−26.9, +13.3, +13.6 in IMR/TMR/OER); proprietary VLMs are directionally closer but add 9.2 OER points. AA reveals a different family split. Humans change by (+25.4, −25.4, 0.0), open VLMs by (+5.7, −12.8, +7.1), and proprietary VLMs by (+34.0, −44.9, +11.0). Opposite assertions reduce TMR for 10 of 12 VLMs; panel (d) shows that open-model TMR falls from 80.6% to 67.8%, whereas proprietary-model TMR falls from 49.1% to 4.2%. Proprietary systems thus exhibit stronger compatibility sensitivity, but their OER increase shows that rejecting the text is not equivalent to recovering the visual reference. These outcome-unfiltered contrasts diagnose condition sensitivity rather than self-retention from image-only judgments.

Diagnostic Reversibility Study

We next ask whether Visual and Text decisions are internally distinguishable and, if so, whether human-discordant Text decisions can be reversed. For an input $x = (I, T)$, let y_v be the human-validated visual reference, y_t the text target, $\hat{y}_m^0$ backbone m 's baseline prediction, and $p_H(y_v | I, T)$ the share of five human judgments that retain y_v . Here p_H refers only to the full-coverage longitudinal annotation panel. The independent split-panel study is not used for intervention-cohort selection because its primary incomplete-block cells intentionally receive one joint response each. The four intervention methods are evaluated on the model-specific oracle cohort

$$\mathcal{T}_{H,m} = \{x \in \mathcal{D}_{\text{full}}^- : p_H(y_v | I, T) \geq 4/5 \wedge \hat{y}_m^0 = y_t\}.$$

Thus, at least four of five humans preserve the visual emotion while the model follows the interfering text. Selection uses both human responses and the baseline output, so this is a post-hoc reversibility diagnostic rather than a deployable detector. We accordingly do not reuse the selected panel's mechanically bounded retention rate as a human baseline. For the post-intervention prediction $\hat{y}_m^1$, we report

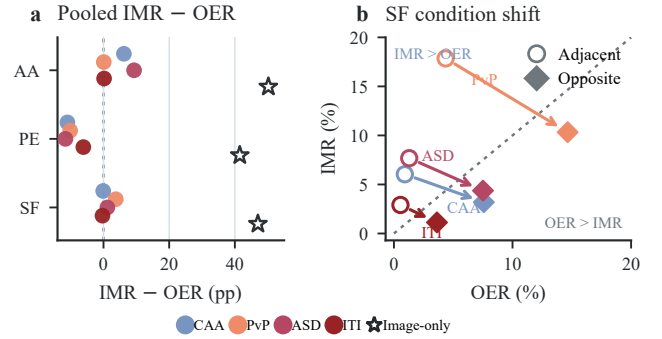


Figure 5: **Text removal is more reversible than activation steering, whose IMR-OER balance remains function- and condition-dependent.** (a) Pooled balance for four activation methods and image-only fallback. (b) Each arrow traces one method from SF-Adjacent to SF-Opposite; the balance reverses sign in all 16 backbone-method pairs. Rates for backbone m are conditional on $\mathcal{T}_{H,m}$.

IMR, TMR, and OER conditional on $\mathcal{T}_{H,m}$, together with $\Delta_{I-O} = \text{IMR} - \text{OER}$. A positive balance means that returns to y_v outnumber redirections to a third emotion.

Internal readability. On Qwen3-VL-8B, matched Visual-Text pairs across AA, PE, and SF develop a sustained answer-onset margin separation from layer 17, reaching 22.04 [19.88, 24.03] at the final layer (Fig. 4(a,b)). An image-disjoint layer-23 readout reaches 0.921 AUROC overall and remains strong within AA (0.947) and PE (0.965), but is weaker within SF (0.668 [0.404, 0.875]). Transfer is asymmetric: SF-to-AA/PE reaches 0.848/0.875, whereas AA/PE-to-SF reaches 0.713/0.776. This indicates a shared component but greater SF heterogeneity, not a universal redirection axis. Matching fixes function, subtask, and both emotion labels; grouped splits keep repeated images together. Label-only controls equal 0.500, and within-stratum permutation gives 0.432 AUROC [0.367, 0.531].

Attention reallocation. On PE, we reallocate half of the answer-onset attention mass from description to image to-

Class	Method	AA			PE			SF			$\Delta\text{Same} \uparrow$	$\Delta\text{Align} \uparrow$
		IMR $\uparrow$	TMR $\downarrow$	OER $\downarrow$	IMR $\uparrow$	TMR $\downarrow$	OER $\downarrow$	IMR $\uparrow$	TMR $\downarrow$	OER $\downarrow$		
Input-level	No intervention	0.00	100.00	0.00	0.00	100.00	0.00	0.00	100.00	0.00	–	–
	Image-only fallback	71.35	7.50	21.15	70.12	1.26	28.62	69.33	8.36	22.31	-24.50	-10.17
Activation	CAA	9.38	87.45	3.17	3.09	82.83	14.08	4.48	90.96	4.57	-0.39	+1.57
	PvP	17.76	64.56	17.68	11.01	67.83	21.16	13.73	76.27	9.99	-0.64	+2.27
	ASD	14.57	80.17	5.26	4.95	78.51	16.54	5.87	89.44	4.69	-0.78	+1.44
	ITI	0.26	99.65	0.08	0.80	92.26	6.94	1.93	95.83	2.24	+0.16	<u>+1.68</u>

Table 4: **Recovery and preservation are distinct (%)**. Pooled rates use model-specific $\mathcal{T}_{H,m}$; $\Delta\text{Same}/\Delta\text{Align}$ are four-backbone macro changes on AA-Same and PE-Aligned controls. Arrows indicate favorable directions; bold/underlined mark the best/second-best activation interventions. Input-level controls are not ranked.

kens. Recovery remains negligible (mean IMR 0.40%; best head 2.15%). Shifted mass does not predict discrete recovery ($\rho = -0.023, p = 0.823$); its margin association is analyzed in the supplementary material.

Intervention protocol. Across Qwen2.5-VL-7B, Qwen3-VL-8B, LLaVA-NeXT-8B, and InternVL3.5-8B, we compare image-only fallback with CAA (Rimsky et al. 2024), PvP (Golovanevsky et al. 2025), ASD (Su et al. 2025), and ITI (Li et al. 2023). Within each backbone, all methods use the same $\mathcal{T}_{H,m}$; pooled rates therefore do not rank backbones. Separate AA-Same and aligned controls measure collateral effects. All data-dependent configuration for the four methods uses a complete, outcome-unfiltered development set of 400 source images. This set is source-image-disjoint from the 1,600-image final benchmark, which is never used for method selection or tuning. Implementation details and cell-level results are in the supplementary material.

Reversibility gap. Figure 5(a) separates text-removal reversibility from activation steering. Image-only fallback reaches 69.33–71.35% IMR and yields AA/PE/SF IMR–OER balances of +50.20 [46.99, 53.19], +41.49 [37.17, 45.96], and +47.02 [43.67, 50.34] points, respectively, although 21.15–28.62% of outputs still enter OER. Activation methods remain much closer to the equal-return boundary: balances range from +0.08 to +9.31 on AA, are uniformly negative on PE (–11.59 to –6.14), and range from –0.31 to +3.74 on SF. Text removal therefore provides a diagnostic ceiling that none of the activation methods approaches.

Table 4 clarifies where the remaining outputs go. PvP attains the highest IMR and lowest TMR in all three functions, but also the highest OER (17.68%, 21.16%, and 9.99%). Conversely, ITI minimizes OER yet recovers only 0.26–1.93% of visual references. Thus, reducing text following can either restore y_v or redirect predictions elsewhere; TMR reduction alone is not correction.

Relation-specific recovery. The pooled SF result further hides a systematic relation effect (Fig. 5(b)). From Adjacent to Opposite, every method moves toward lower IMR and higher OER: pooled IMR–OER balance changes from +2.36–+13.49 points on Adjacent to –4.39––2.50 points on Opposite. The same sign reversal holds in all 16 backbone–method cells, showing that it is not driven by one architecture. Reversibility therefore depends not only on the

intervention, but also on the affective relation it is applied to.

Preservation and collateral effects. Activation steering largely preserves consistent controls. Across 16 method–backbone cells, AA-Same remains 96.19%–99.81% (mean change –0.42 points), while aligned accuracy changes by +1.74 points on average and is non-decreasing in 12 cells. ITI is non-negative on both controls for every backbone but nearly inert on the oracle cohorts. Effects remain backbone dependent: Qwen3-VL-8B loses up to 3.06 AA-Same and 0.81 aligned points. Without cell-level intervals, we interpret small gains as preservation rather than improvement.

Image-only fallback instead lowers AA-Same and PE-Aligned accuracy by 24.50 and 10.17 points. In a Qwen3-VL-8B label-logit audit, visual-first and confidence-gated prompting recover 66.97/66.47% of PE and 66.56/65.39% of SF cases. Gating reduces damage to baseline Visual outputs from 0.74% to 0.50% on PE and from 3.07% to 1.48% on SF, at a 1.37-point aligned-accuracy cost. Because this audit uses a different decoder, routing details remain supplementary. Input removal thus offers reversibility at preservation cost, whereas activation steering preserves controls but recovers few visual references; neither achieves both.

Conclusion

AC-Bench provides a controlled evaluation of how competing language redirects visual-emotion judgments. Across AA, PE, and SF, the independent split-panel comparison shows that the evaluated VLM suite retains the visual reference less often and follows the text target more often than humans on the same conflict cells. Mechanistic analyses further reveal separable Visual and Text decision states, yet current activation interventions reverse few human-discordant shifts; text removal recovers substantially more cases but damages aligned controls. These results frame affective robustness as selective cross-modal arbitration: models must reject visually unsupported interpretations without discarding useful linguistic context.

AI Use Statement. AI assisted only with language polishing and coding.

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